

All orders asymptotics for square lonesum decomposable matrices

Report 132

October 2, 2026

Abstract

Let a_n count square binary matrices that split, after independent row and column permutations, into lonesum blocks and zero rows and columns. Starting from Kamano's exact bivariate generating function, we prove a complete expansion in powers of $n^{-1/2}$, with an explicit leading constant and first two correction coefficients. The proof treats the anisotropic saddle and all complementary arcs. We also give a coefficient algorithm, exact integer enumeration, and all-orders inverse threshold brackets that keep rounding uncertainty inside the ceiling. The results concern OEIS A299907, equivalently induced- P_5 -free bipartite graphs on two fixed labelled shores of size n .

[write] **Status and trust boundary, added 6 October 2026, batch 108.** This is bundle Report 132 of an external AI-assisted research session, filed in ProveIt as the single-source research report **a299907-lonesum-decomposable-matrices** (provenance, the sources as the write read them, what was checked, relation to the repository, the collected non-claims and the reading conventions are in Section 1.1). It is unrefereed and not formalized; its author line reads "Report 132" and its PDF author field is empty. *What rests on what.* Kamano's generating function (1) and the exact enumeration (15) are cited prior results (the latter is his Proposition 4.1 summed over the decomposition order; Remark 4). Theorem 1 is the source's own; its proof (Sections 3–4, with Lemma 1 for the global contour) uses nothing but (1), and its remainder constants are existential. The text gives the steps but not the algebra behind the printed γ , A , B , P_1 , P_2 , c_1 and c_2 ; the delivered computer-algebra scripts reproduce them, and so does the write's own expansion, a separate program written from (9). Section 5 rests on Theorem 1 and on the elementary doubling $a_{n+1} \geq 2a_n$. The finite checks of Section 6 prove nothing asymptotic. The write found no error in the source. It adds, with proofs, Remark 1 (lonesum decomposability is a property of the connected components, each a chain graph, and the forbidden configurations are exactly the induced paths P_5 , both directions), Remark 4 (the three OEIS entries quoted; the entry's Mathematica double sum is (15); a misprinted value in the example table of A299906), Remark 3 (Section 5 against the transseries volume, statement by statement), Remark 2 (the third coefficient c_3 , which the source computed but did not claim) and questions in Section 7. Nothing was submitted to the OEIS.

1 The counted objects and prior results

A binary matrix is *lonesum* if its row-sum and column-sum vectors determine it uniquely among binary matrices of the same dimensions. A matrix is *lonesum decomposable* if independent permutations of its rows and columns turn it into a block diagonal matrix with lonesum blocks having no zero row or column, together with optional zero rows and columns outside those blocks. We count the original labelled matrices, not equivalence classes under those permutations. Write $D(m, n)$ for this count and $a_n = D(n, n)$, including $a_0 = 1$ and the all-zero matrix in every dimension. The diagonal begins

1, 2, 16, 344, 13528, 833432, 73871416

and is OEIS A299907; the rectangular array is A299906 [2, 3].

Kamano proves uniqueness of the nonzero block decomposition and the exact exponential generating function [1, Proposition 1.2; Theorem 3.1, equation (9)]

$$F(x, y) = \sum_{m, n \geq 0} D(m, n) \frac{x^m y^n}{m! n!} = \exp \left(x + y + \frac{1}{1 - (e^x - 1)(e^y - 1)} - 1 \right). \quad (1)$$

The constant -1 is important: it ensures $F(0, 0) = 1$.

There is also a useful graph interpretation. Regard a matrix as the bipartite adjacency matrix on fixed labelled row and column shores. Kamano's forbidden submatrix is

$$U = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix},$$

up to row and column permutations and transposition [1, Theorem 2.1]. Its graph is exactly the induced path $y_2 - x_1 - y_1 - x_2 - y_3$ on five vertices. Thus the forbidden configurations are precisely induced copies of P_5 . Each nontrivial connected component is a chain (or Ferrers) graph. The theorem below consequently enumerates induced- P_5 -free bipartite graphs with this prescribed balanced bipartition; it does not count graphs up to isomorphism or graphs with an unspecified bipartition.

Remark 1 ([write] Components, chain graphs and induced paths, 6 October 2026). Two sentences of the preceding paragraph are stated without proof: that each nontrivial connected component is a chain graph, and that the forbidden configurations are *precisely* the induced copies of P_5 (the paragraph checks one direction, that the graph of U is an induced P_5). Both follow from Kamano's results. For an $m \times n$ binary matrix M let $G(M)$ be its bipartite graph on the labelled row shore $\{x_1, \dots, x_m\}$ and column shore $\{y_1, \dots, y_n\}$, with $x_i y_j$ an edge when $M_{ij} = 1$.

(a) *A lonesum matrix without zero rows or columns has a connected chain graph.* By Ryser's theorem in Kamano's form [1, Proposition 1.1], a binary matrix is lonesum if and only if it contains neither $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ nor $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ as a submatrix. Then any two row supports R_1, R_2 are comparable under inclusion: otherwise columns $c \in R_1 \setminus R_2$ and $c' \in R_2 \setminus R_1$ would give one of the two forbidden submatrices in rows 1, 2 and columns c, c' . Nested neighbourhoods on the row shore are the definition of a chain (difference, Ferrers) graph. If there is no zero column, every column has a 1 in some row, whose support lies in a row support of largest size; so that row meets every column and, with no zero row, the graph is connected.

(b) *M is lonesum decomposable if and only if every connected component of $G(M)$ with at least one edge has a lonesum biadjacency submatrix.* If M is lonesum decomposable, the blocks of its decomposition occupy disjoint sets of rows and columns and contain every 1; by (a) each block is connected, so the blocks are exactly the components with an edge. Conversely, listing those components one after another, then the isolated rows and columns, is a row and column permutation to the required block form. In particular every nontrivial component is a connected chain graph, as the text says.

(c) *The forbidden configurations are exactly the induced P_5 .* Kamano's set $\mathcal{N}$ [1, Section 2] consists of the twelve matrices obtained from U and its transpose by permuting rows and columns, and M is lonesum decomposable if and only if it contains no member of $\mathcal{N}$ as a submatrix [1, Theorem 2.1]. A 2×3 submatrix in rows i_1, i_2 and columns j_1, j_2, j_3 is a row and column permutation of U if and only if each of its two rows has exactly two ones and the two rows share exactly one column (two distinct 2-subsets of a 3-set meet in one element, and a column permutation carries any such pair to $\{1, 2\}, \{1, 3\}$). In $G(M)$ this says that x_{i_1} and x_{i_2} have a common neighbour y and

private neighbours y' , y'' among the three columns, with $x_{i_1}y''$ and $x_{i_2}y'$ absent: $y' - x_{i_1} - y - x_{i_2} - y''$ is an induced path, the other pairs of its five vertices lying on one shore. Conversely, an induced P_5 with vertices $v_1, \dots, v_5$ in path order alternates shores, so v_2, v_4 lie on one shore and v_1, v_3, v_5 on the other; its edges are $v_1v_2, v_2v_3, v_3v_4, v_4v_5$ and the pairs v_1v_4, v_2v_5 are non-edges. The submatrix on rows and columns $\{v_2, v_4\} \times \{v_1, v_3, v_5\}$ (or its transpose, when v_2, v_4 are columns) is therefore a permutation of U (or of its transpose). Hence M is lonesum decomposable if and only if $G(M)$, with both shores fixed, has no induced P_5 : Kamano's Theorem 2.1 restated. The delivered checker's exhaustive count through $n = 3$ tests exactly the criterion of this paragraph.

(d) *A check.* The write enumerated every $m \times n$ binary matrix for $1 \leq m, n \leq 4$ and decided lonesum decomposability three ways: from the definition through (b), with lonesumness tested directly as uniqueness among all matrices with the same row and column sums (no use of Ryser's theorem); by the induced- P_5 criterion of (c); and by the rectangular form $D(m, n) = \sum_k k! B_k S(m+1, k+1) S(n+1, k+1)$ of (15). The three agree for all sixteen shapes, for instance $D(2, 3) = 58$, $D(3, 4) = 1786$ and $D(4, 4) = 13528$.

Prior-art boundary. Equation (1) and the exact enumeration are prior results. Ordinary lonesum matrices have generating function $e^{x+y}/(1 - (e^x - 1)(e^y - 1))$; their asymptotics, including higher-order extensions, are already treated by Khera, Lundberg and Melczer [5]. Their use of the symbol D for an ordinary-lonesum variant is unrelated to $D(m, n)$ here. The present exponential of a pole has different saddle scales. Related restricted decomposable classes are studied by Bényi and Ramírez [6]. A scoped search of the inspected OEIS entries and accessible primary literature located no balanced-shore asymptotic equivalent for this decomposable class. The full text of Khera's 2021 dissertation was not available in that search [7]. This is a limitation on priority assessment, not a claim of worldwide novelty or the resolution of a named open conjecture.

1.1 Provenance, sources and reading conventions

[write] **Added 6 October 2026, batch 108.** *Provenance.* This report is Report 132 of the session bundle that ProveIt's docs/incoming intake processed as batch 108: the archive `A299907_All_Orders_Matrix_Asymptotics_and_Inversion_Source.zip` (369,597 bytes, SHA-256 `ad4d45c02c09...b16feeaa1362`, 18 files in the wrapper directory `Report132_bundle`), arrival commit `60f54ea06`, placed in `602e5bd0f`. The text is the delivered `Report132.tex` (334 lines, dated 2 October 2026), shipped as `article.tex`. The delivered eight-page PDF and the integrity manifest `MANIFEST.json` (17 entries, verified at placement) are not shipped, and the delivered README is replaced by the report README; all three are retrievable from the arrival commit. The package records no ProveIt commit; its `sources.md` mentions a "bounded repository-topic review" without naming a repository or commit. It contains no "prepared for private review" line and no personal data. It is a single-source report, so no merge choice was made. The write prefixed the 17 delivered labels with `1sd:` and updated the 14 references to them; it added the notes marked [write], this subsection, Remarks 1–4, the label and the added part of Section 7, and nothing else. No statement, proof or number of the source was changed; the source has no remark environment, so the added remarks carry their own numbers, and every theorem, lemma, section and equation keeps its delivered number.

The sources, as the write read them (6 October 2026).

- Kamano [1], arXiv:1701.07157v1: Propositions 1.1 (Ryser's characterization), 1.2 (uniqueness of the decomposition matrix) and 4.1 (the Stirling formula for $D_k(m, n)$), Section 2 with the set $\mathcal{N}$, Theorem 2.1 (the forbidden submatrices) and Theorem 3.1 with equation (9), which is (1) after $e^x + e^y - e^{x+y} = 1 - (e^x - 1)(e^y - 1)$; it is on page 7 of the arXiv PDF. The journal version was not read.
- Khera, Lundberg and Melczer [5], arXiv:1912.08850v2: Remark 2 (higher-order terms for square lonesum matrices) and their $D_{n,k}$, lonesum matrices without zero rows or columns (not Kamano's

$D_1(m, n)$, which allows zero lines outside its one block). Bényi and Ramírez [6], arXiv:1804.03949v1, Section 4.3 and Theorem 11 (restricted lonesum decomposable matrices), as cited.

- OEIS A299907, A299906 and A000262, live entries fetched on 6 October 2026 (Remark 4). The source’s `sources.md` says that its own direct OEIS requests “failed” during final preparation and that the identification and the fixture transcription were inherited from an earlier review; the write’s fetch confirms both.
- The transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`: `p0:prop:factorial-core`, `p0:thm:core-reversion`, `p0:rem:core-instances`, `p0:def:three-inverses`, `p0:thm:staircase`, `p0:thm:lambert-centered`, `plt:def:lw-monomial-log-datum` and `plt:thm:lw-template`, read for Remark 3.
- Not read by the write: Khera’s dissertation [7] (the source did not read it either) and the journal texts behind the arXiv versions above.

What was checked at intake. The batch-108 dossier read the manuscript in full and found no error; it checked by hand the induced- P_5 reading of U , the constant C from (3) and the $-\frac{1}{4}\log n$ term of (12), and reran the delivered checker, both coefficient programs and the $n = 100$ diagnostic on a copy. The write rechecked every proof line by line and found no error. Independently of the delivered programs it re-derived, with its own series code, the Laurent expansion (9) through t^4 (the cancellation of the t^{-4} , t^{-3} and t^{-1} terms, β , γ , A , B , P_1 , P_2) and from it c_1 and c_2 exactly; it checked the two δ -expansions and r_n of Section 3, and at the exact saddle for $n = 10^4, 10^6, 10^8$ the eigenvalue identity $B_{11} - B_{12} = n - r_n^2 s/D^2$ and both limits of (7); it verified the identity used in the proof of Lemma 1 and its case analysis, the Jacobian $2t^5$, the prefactor, the parity argument, the three cancellation identities of Section 5 (also as Remark 3(b)), the doubling injection and the derivations of (14) and (15); it recomputed the table of Section 6 for $n = 100, 400, 800$ from exact integers (the row $n = 3200$ equals the delivered diagnostics file) and the brute-force counts of Remark 1(d). Finite computations are observations, not certificates.

Relation to the repository. Before batch 108 no file of the repository named A299907, A299906 or lonesum decomposable matrices; A000262 occurred only in `congruences-and-valuations/a321941-asymptotic-coefficient-integrality`, as an unrelated tool. The nearest report is `a122399-surjection-diagonal` (the diagonal of $1/(1 - ue^y(e^x - 1))$), a pole rather than the exponential of a pole, with the same $(n!)^2$ normalization and Stirling machinery; it credits Khera, Lundberg and Melczer; `a089479-fixed-permanent-matrices`, of the same batch, mentions a lonesum-decomposable report as a different model. No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. Generic ingredients are formalized in the FabiusFunction library (`Analysis/FabiusFunction/Lean/FabiusFunction/`): the Lah numbers `Fabius.lahNumber` with their column generating function `Fabius.X_mul_mkOne_pow`, $(t/(1-t))^k = \sum_n k! L(n, k) t^n / n!$, from which $\sum_k L(n, k) = B_n$ (A000262) follows by summing over k ; the vertical recurrence `Fabius.stirlingSecond_succ_succ_eq_sum`; and the generic rounding facts `Fabius.staircase_ceil` and `Fabius.staircase_separation`. None of them is a statement of this report.

[write] **What is not claimed, collected from the source.** The constants in the remainders of Theorem 1 and in the inverse brackets, and the onset thresholds, are existential; no certified finite- n error bound or finite- T bracket is given, and numerical tests do not supply one. No worldwide novelty or priority is claimed (“a limitation on priority assessment, not a claim of worldwide novelty or the resolution of a named open conjecture”); the source search was bounded, Khera’s dissertation was not inspected, and the exact generating function and enumeration are Kamano’s. The graph statement fixes the labelled bipartition: it does not count graphs up to isomorphism or with an unspecified bipartition. $\lceil X_1 \rceil$ is not a valid rounding rule; the error must stay inside the ceiling. No central or local limit theorem for the number of blocks is asserted, and optimal truncation and exponentially small terms are not identified. The diagnostics file stores c_3 , which the source does not claim. The finite checks and the quality-assurance procedure check reproducibility and named finite identities only; they do not mechanize the analytic proof; integrity hashes are not authorship authentication; PDF byte identity holds only in the delivering TeX environment. The write adds: the value of c_3 in Remark 2 is a computer-algebra output of the delivered recipe, and the comparisons in Remark 3 classify statements; they prove nothing new about a_n .

Reading conventions (letters the source reuses, with the tempting false readings; no symbol of the source was renamed; symbols of the transseries volume carry the subscript “vol”).

Symbol	Meaning here (where)	Not to be read as
$D(m, n), a_n$	lonesum decomposable $m \times n$ matrices; $a_n = D(n, n)$	lonesum matrices (Kamano’s $L(m, n)$); KLM’s $D_{n,k}$; Kamano’s $D_k(m, n)$ (order k); A299906’s $T(n, k) = D(n, k)$
$D; D_1, D_2$	$1 - q^2$ (Section 3); inverse corrections (Section 5)	the count $D(m, n)$
$A(z); A$	$e^z - 1$; the Gaussian constant $4\sqrt{2\ell}$	each other
$B_{11}, B_{12}; B;$ B_k	Hessian entries; $1 - \ell$; A000262 (Section 6)	Bell numbers; KLM’s poly-Bernoulli $B_{n,k}$
k	$A(r)^2 - \rho$ (proof of Lemma 1); $\log(2\pi C)$ (Section 5); an index (Sections 4, 6)	Kamano’s decomposition order
$c_j; c, c_0, \eta$	expansion coefficients, $c_0 = 1$; the constants of Lemma 1, whose c_0 is a gap	each other
$d; d_j$	$1 - A(r)^2$ (Lemma 1); coefficients of $\log(1 + \sum c_j z^j)$	d_{vol} of <code>p0:prop:factorial-core</code>
h	$(x - y)/2$ (Section 4); $\log(x/\ell)$ (Section 5)	$h_{\text{vol}}(E)$ of <code>p0:thm:core-reversion</code>
$x, y; m$	generating-function variables, but in Section 5 the Lambert root and the real variable of Φ_M ; the row count, and $(x + y)/2$ in Section 4	
ℓ, L, T	$\log 2$; $\log T$; a threshold	$\ell_{\text{vol}} = \log X, L_{\text{vol}} = \log Z$ of <code>plt:thm:lw-template</code> ; A299906’s T
$t, U, V; u, v$	$n^{-1/4}$ and scaled angles; common and opposing angles	$t_{\text{vol}}, v_{\text{vol}}$
β, γ	$\sqrt{2/\ell}; 2\ell - \frac{5}{8} + \frac{1}{8\ell}$	Euler’s constant; β_{vol}
$P; P_j, Q_j, \mathbb{E}$	$A(re^{i\theta})A(re^{i\phi})$ (Lemma 1); expansion polynomials; the Gaussian functional	
$N(T), W$	$\min\{n : a_n \geq T\}$; the branch W_0	equal to the staircase N_* of <code>p0:def:three-inverses</code> here (Remark 3)
$S(n, k)$	Stirling numbers of the second kind	

2 Main expansion

Put $\ell = \log 2$ and $\beta = \sqrt{2/\ell}$.

Theorem 1. *For each fixed integer $M \geq 0$, there are explicitly computable real constants c_j such that*

$$a_n = (n!)^2 C \ell^{-2n} n^{-5/4} e^{\beta\sqrt{n}} \left(\sum_{j=0}^M c_j n^{-j/2} + O_M(n^{-(M+1)/2}) \right), \quad (2)$$

where $c_0 = 1$ and

$$C = \frac{\exp(-5/8 + 1/(8\ell))}{\pi(2\ell)^{1/4}\sqrt{1-\ell}}, \quad (3)$$

$$c_1 = -\frac{\sqrt{2}(167\ell^3 - 143\ell^2 - \ell + 1)}{192\ell^{3/2}(\ell - 1)}, \quad (4)$$

$$c_2 = \frac{65041\ell^6 - 108866\ell^5 + 42099\ell^4 + 1724\ell^3 - 573\ell^2 - 2\ell + 1}{36864\ell^3(\ell - 1)^2}. \quad (5)$$

Numerically $C = 0.3394739586934123704\dots$, $c_1 = -0.5317063378238041150\dots$, and $c_2 = -0.1635701301266242142\dots$

3 The saddle and exclusion of other arcs

Write $g = \log F$ and $A(z) = e^z - 1$. For any $0 < r < \ell$, the closed bidisc $|x|, |y| \leq r$ lies in the domain of analyticity of F : indeed $|A(x)A(y)| \leq A(r)^2 < 1$. The positive symmetric saddle is the unique solution of

$$n = r + \frac{re^r(e^r - 1)}{[1 - (e^r - 1)^2]^2}. \quad (6)$$

The right side increases strictly from 0 to infinity: it is a power series with nonnegative coefficients and a positive linear coefficient. With $\delta = \ell - r$,

$$1 - (e^r - 1)^2 = 4\delta - 6\delta^2 + \frac{14}{3}\delta^3 + O(\delta^4), \quad \frac{1}{1 - (e^r - 1)^2} = \frac{1}{4\delta} + \frac{3}{8} + \frac{13}{48}\delta + O(\delta^2).$$

The right side of (6) equals $\ell/(8\delta^2) - 1/(8\delta) + 83\ell/96 + O(\delta)$, giving

$$r_n = \ell - \sqrt{\ell/8} n^{-1/2} + \frac{1}{16} n^{-1} + O(n^{-3/2}).$$

Let $s = e^{r_n}$, $q = s - 1$, and $D = 1 - q^2$. The angular Hessian has diagonal entry

$$B_{11} = n + r_n^2 \left(\frac{sq}{D^2} + \frac{2s^2 q^2}{D^3} \right)$$

and off-diagonal entry

$$B_{12} = r_n^2 \left(\frac{s^2}{D^2} + \frac{2s^2 q^2}{D^3} \right).$$

Its eigenvalues satisfy

$$\lambda_+ \sim 4\sqrt{2\ell} n^{3/2}, \quad \lambda_- = n - \frac{r_n^2 s}{D^2} \sim (1 - \ell)n. \quad (7)$$

Thus the common-angle width is $n^{-3/4}$ and the opposing-angle width is $n^{-1/2}$.

Here is the required global contour estimate; a local Hessian calculation alone would not justify the expansion.

Lemma 1. *Uniformly for $r < \ell$ sufficiently near ℓ , put $d = 1 - A(r)^2$, $\theta = u + v$, $\phi = u - v$, and $P = A(re^{i\theta})A(re^{i\phi})$. There are fixed positive constants c, η such that for $|\theta|, |\phi| \leq \eta$,*

$$\frac{1}{d} - \Re \frac{1}{1 - P} \geq c \min \left(\frac{v^2}{d^2} + \frac{u^2}{d^3}, \frac{1}{d} \right). \quad (8)$$

On the remainder of $[-\pi, \pi]^2$, after deleting this neighborhood, the left side is at least c/d .

Proof. Near zero write $P = \rho e^{i\omega}$, with the continuous argument vanishing at zero. For $\psi_r(\theta) = \arg A(re^{i\theta})$, its derivative at zero is $\mu_r = re^r/(e^r - 1) > 0$. The log-modulus curvature is

$$\sigma_r^2 = \frac{re^r}{e^r - 1} - \frac{r^2 e^r}{(e^r - 1)^2} \longrightarrow 2\ell(1 - \ell) > 0.$$

Hence, uniformly in r , $A(r)^2 - \rho \asymp u^2 + v^2$. Also $\omega = \psi_r(u + v) + \psi_r(u - v)$. Since ψ_r is odd, this equals u times a continuous even function of u whose value at $(u, v) = (0, 0)$ is $2\mu_r$. Shrinking the neighborhood gives $|\omega| \asymp |u|$.

The exact identity

$$\frac{1}{1 - \rho} - \Re \frac{1}{1 - \rho e^{i\omega}} = \frac{\rho(1 + \rho)(1 - \cos \omega)}{(1 - \rho)((1 - \rho)^2 + 2\rho(1 - \cos \omega))}$$

now proves the claim. In detail let $k = A(r)^2 - \rho$, so $1 - \rho = d + k$. The difference between $1/d$ and $1/(1 - \rho)$ is $k/[d(d + k)]$. If $k \geq d$, this is at least $1/(2d)$. If $k < d$, the sum of the two displayed differences is bounded below by a positive constant times

$$\frac{u^2 + v^2}{d^2} + \frac{u^2}{d(d^2 + u^2)},$$

which implies (8).

Away from zero, the triangle inequality for $A(re^{i\theta}) = \sum_{j \geq 1} r^j e^{ij\theta}/j!$ is strict unless $\theta = 0 \pmod{2\pi}$ (the terms $j = 1, 2$ suffice for strictness). Compactness, with r restricted to a closed interval ending at ℓ , gives a fixed gap $\rho \leq A(r)^2 - c_0$ outside the neighborhood. The reciprocal is then uniformly bounded, while $1/d \rightarrow \infty$. $\square$

Since $\Re(re^{i\theta} + re^{i\phi}) \leq 2r$, the lemma also bounds the loss of the real part of g . In particular the contribution outside

$$|u| \leq n^{-3/4} \log n, \quad |v| \leq n^{-1/2} \log n$$

is smaller than the central exponential scale by $\exp(-c \log^2 n)$, even after accounting for the polynomial central area. No competing point of the Cauchy torus contributes.

4 Local expansion and coefficient recipe

For extraction it is simpler to use the explicit radius $r = \ell - \alpha t^2$, where $\alpha = \sqrt{\ell/8}$ and $t = n^{-1/4}$, rather than to expand the exact saddle. Set

$$\theta = t^3 U + t^2 V, \quad \phi = t^3 U - t^2 V, \quad m = re^{it^3 U} \cos(t^2 V), \quad h = ire^{it^3 U} \sin(t^2 V).$$

Then $x = m + h$, $y = m - h$, and

$$H = 1 - A(x)A(y) = 2e^m \cosh h - e^{2m}.$$

Cauchy's formula reads

$$\frac{a_n}{(n!)^2} = \frac{r^{-2n}}{(2\pi)^2} \int_{[-\pi, \pi]^2} \exp\{g(re^{i\theta}, re^{i\phi}) - in(\theta + \phi)\} d\theta d\phi.$$

Define the following formal Laurent series, keeping $\ell = \log 2$ so $e^\ell = 2$:

$$\mathcal{E}(t, U, V) = 2m + H^{-1} - 1 - 2t^{-4} \log(r/\ell) - 2iU/t = \beta t^{-2} + \gamma - AU^2 - BV^2 + \sum_{j \geq 1} P_j(U, V) t^j, \quad (9)$$

where

$$\gamma = 2\ell - \frac{5}{8} + \frac{1}{8\ell}, \quad A = 4\sqrt{2\ell}, \quad B = 1 - \ell.$$

The t^{-1} term cancels. Each P_j is a polynomial in U, V . In particular

$$\begin{aligned} P_1 &= -iU \left(16\ell U^2 + 4\sqrt{2\ell}(1 - \ell)V^2 + \frac{1}{\sqrt{2\ell}} \right), \\ P_2 &= \sqrt{2\ell} \left(32\ell U^4 + (1 - \ell)^2 V^4 + \left(\frac{\ell - 1}{2} + \frac{1}{4\ell} \right) V^2 - \frac{83}{192} + \frac{1}{48\ell^2} \right) \\ &\quad + 24\ell(1 - \ell)U^2V^2 + 3U^2. \end{aligned}$$

Let $Q_0 = 1$ and

$$Q_j = \frac{1}{j} \sum_{k=1}^j k P_k Q_{j-k}. \quad (10)$$

These are the coefficients of $\exp(\sum_{j \geq 1} P_j t^j)$. Let $\mathbb{E}$ denote the product Gaussian functional with density $\sqrt{AB} e^{-AU^2 - BV^2} / \pi$. Then the all-orders recipe is

$$c_j = \mathbb{E} Q_{2j}. \quad (11)$$

For monomials the functional is explicit:

$$\mathbb{E}(U^{2p} V^{2q}) = \frac{(2p-1)!!(2q-1)!!}{(2A)^p (2B)^q},$$

and odd moments vanish. Thus each coefficient involves only finitely many exact algebraic operations over $\mathbb{Q}(\sqrt{\ell}, \sqrt{2})$. For example $c_1 = \mathbb{E}(P_2 + P_1^2/2)$ gives (4); continuing the same recurrence gives (5).

For completeness, this formal manipulation is an asymptotic expansion with controlled remainder. On the central logarithmic box, $H/(4\alpha t^2) = 1 + O(t|U| + t^2(1 + V^2))$, so its reciprocal and all other factors have ordinary convergent Taylor expansions after removing the displayed Laurent terms. For each fixed order, Taylor remainders are bounded by a fixed power of t times a polynomial in $|U| + |V|$. For a fixed constant $K > 0$, the exponential of the perturbation is bounded in modulus by $K \exp((AU^2 + BV^2)/2)$ on this box for sufficiently small t . Integrating the remainder against the remaining Gaussian removes any artificial logarithmic factors. Lemma 1 applies equally to this explicit radius, since $1 - A(r)^2 \asymp t^2$, and makes the complement smaller than every power of t .

Moreover $\mathcal{E}(-t, U, V) = \mathcal{E}(t, -U, V)$, so Q_j is odd in U when j is odd. All odd powers integrate to zero. Expanding through degree $2M + 1$ therefore leaves $O(t^{2M+2}) = O(n^{-(M+1)/2})$. Finally the Jacobian is $d\theta d\phi = 2t^5 dU dV$, and the central Gaussian integral is $\pi/\sqrt{AB}$. Its total prefactor is

$$\frac{e^\gamma}{2\pi\sqrt{AB}} = \frac{e^\gamma}{4\pi(2\ell)^{1/4}\sqrt{1-\ell}} = C,$$

because $e^{2\ell} = 4$. This proves Theorem 1.

Remark 2 ([write] The third coefficient, 6 October 2026). The delivered diagnostics file records $c_3 = -0.26674650591076156210735\dots$, and the delivery README says that the article does not claim it. Run by the write on a copy (about ten seconds), the delivered program `derive_general.py`

at order 3, which evaluates (11) by computer algebra in the variable $z = \sqrt{\ell/8}$, returns after the substitution $z^2 = \ell/8$

$$c_3 = -\frac{\sqrt{2} N_3(\ell)}{106168320 \ell^{9/2} (\ell - 1)^3},$$

$$N_3(\ell) = 150570067\ell^9 - 331569921\ell^8 + 185237466\ell^7 + 22417538\ell^6$$

$$- 30193908\ell^5 + 401964\ell^4 + 35446\ell^3 - 9042\ell^2 - 15\ell + 5,$$

numerically $-0.266746505910761562107\dots$, the stored value. This is a computer-algebra output, not a derivation by hand; the write's own expansion (Section 1.1) was taken only to the order of c_2 . The exact counts agree with it: $n^{3/2}(R_n - 1 - c_1 n^{-1/2} - c_2 n^{-1})$ equals -0.2188 , -0.2324 , -0.2422 at $n = 400, 800, 1600$ (the write's computation) and -0.2493 at $n = 3200$ (the diagnostics file), and its differences from c_3 , multiplied by $\sqrt{n}$, are $0.959, 0.972, 0.981, 0.987$, the behaviour expected from $c_4 = 1.0034201765950181\dots$, the value of the recipe at order 4. (Made exact after the independent check of 6 October 2026, which computed c_4 by its own expansion and with the delivered program; the first wording read "the behaviour expected if c_4 is close to 1".) The denominators $192\ell^{3/2}(\ell - 1)$, $36864\ell^3(\ell - 1)^2 = 192^2\ell^3(\ell - 1)^2$ and $106168320\ell^{9/2}(\ell - 1)^3 = 15 \cdot 192^3\ell^{9/2}(\ell - 1)^3$ and the numerator degrees 3, 6, 9 suggest a pattern; it is recorded as Question 5, not claimed.

5 Inversion and integer thresholds

The asymptotic inverse must be distinguished from exact integer rounding. Define

$$N(T) = \min\{n \geq 0 : a_n \geq T\} \quad (T > 1).$$

The sequence obeys $a_{n+1} \geq 2a_n$: adjoining a last row and column that are zero except for a freely chosen bottom-right entry 0 or 1 gives two disjoint injections. The first construction adds isolated row and column vertices; the second adds a separate 1×1 lonesum block. Hence both preserve lonesum decomposability, and deleting the last row and column recovers the input. Thus the sequence is strictly increasing. Writing $k = \log(2\pi C)$, Stirling's formula gives

$$\log a_n = 2n \log \frac{n}{e\ell} + \beta\sqrt{n} - \frac{1}{4} \log n + k + \frac{c_1}{\sqrt{n}} + O(n^{-1}). \quad (12)$$

For $L = \log T$, let

$$x = \frac{L}{2W(L/(2e\ell))}, \quad h = \log(x/\ell),$$

where W is the positive real Lambert function. This x solves $2x \log(x/(e\ell)) = L$. Define

$$D_1 = \frac{\frac{1}{4} \log x - k}{2h} + \frac{\beta^2}{8h^2} - \frac{\beta^2}{8h^3},$$

$$X_1 = x - \frac{\beta\sqrt{x}}{2h} + D_1,$$

$$D_2 = -\frac{c_1 + \beta D_1(\frac{1}{2} - \frac{1}{h}) + \frac{\beta}{8h} + \frac{\beta^3}{24h^3} - \frac{\beta^3}{32h^2}}{2h\sqrt{x}}, \quad X_2 = X_1 + D_2.$$

There are fixed positive K_1, K_2 and sufficiently large T_0 such that, for $T \geq T_0$,

$$\left\lceil X_1 - \frac{K_1}{\sqrt{x}h} \right\rceil \leq N(T) \leq \left\lceil X_1 + \frac{K_1}{\sqrt{x}h} \right\rceil,$$

$$\left\lceil X_2 - \frac{K_2}{xh} \right\rceil \leq N(T) \leq \left\lceil X_2 + \frac{K_2}{xh} \right\rceil. \quad (13)$$

These constants and the onset threshold are existential here; numerical tests do not turn them into certified finite- T error bounds. To verify the orders, expand the right side of (12) at x . The term $-\beta\sqrt{x}/(2h)$ cancels the order- $\sqrt{x}$ term. The displayed D_1 cancels all order-one terms, leaving $O(x^{-1/2})$ in the logarithm. The numerator of D_2 is exactly the remaining coefficient at order $x^{-1/2}$; adding D_2 leaves $O(x^{-1})$. Division by the derivative $2h + O(x^{-1/2})$ gives the stated location scales. Applying increasing upper and lower smooth envelopes to the integer values in (12), as detailed below, gives (13). No interpolation of the unknown exact sequence is being assumed.

In particular one may write $N(T) = \lceil X_1 + O((\sqrt{x}h)^{-1}) \rceil$, provided the error is explicitly kept *inside* the ceiling. There is no unconditional justification for replacing this by $\lceil X_1 \rceil$ for all large T : a threshold may be arbitrarily close to a jump. If the error interval crosses an integer, compare the adjacent exact a_n values.

An all-orders version is equally direct. Define d_j by

$$\log \left(1 + \sum_{j \geq 1} c_j z^j \right) = \sum_{j \geq 1} d_j z^j$$

and set

$$\Phi_M(y) = 2 \log \Gamma(y+1) - 2y \log \ell - \frac{5}{4} \log y + \beta \sqrt{y} + \log C + \sum_{j=1}^M d_j y^{-j/2}.$$

For fixed M , let y_M be the sufficiently large real solution of $\Phi_M(y_M) = \log T$. Then

$$\lceil y_M - E_M \rceil \leq N(T) \leq \lceil y_M + E_M \rceil, \quad E_M = K_M \frac{y_M^{-(M+1)/2}}{\log y_M}$$

for a fixed K_M and all sufficiently large T . To justify the ceiling bracket directly, the expansion gives $|\log a_n - \Phi_M(n)| \leq A_M n^{-(M+1)/2}$ for all sufficiently large integers n . In a fixed relative neighborhood of y_M , the derivative of Φ_M is bounded below by a positive multiple of $\log y_M$. Choose K_M large enough that moving by E_M makes the change in Φ_M dominate this error. Every integer below $y_M - E_M$ then has $a_n < T$, and the integer $\lceil y_M + E_M \rceil$ has $a_n \geq T$. Monotonicity covers smaller integers outside that neighborhood. This proves the bracket without defining an exact real-valued extension of a_n .

The real root can be found by Newton iteration from x . In the relevant neighborhood $\Phi'_M \asymp \log x$ and $\Phi''_M = O(1/x)$, so an iteration with location error e produces error $O(e^2/(x \log x))$. A fixed number of iterations, chosen from the requested order, obtains any prescribed algebraic accuracy. If an approximate numerical root is used in a ceiling bracket, its certified location error must be added to E_M . This is a constructive all-orders inverse, with the unavoidable rounding caveat.

Remark 3 ([write] Section 5 against the transseries volume, 6 October 2026). The volume is **Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/**. Its symbols carry the subscript “vol” where they collide with this report’s. Statement by statement:

- (a) *The Lambert root x is an instance*, verbatim, of `p0:prop:factorial-core`, with $(\kappa_{\text{vol}}, d_{\text{vol}}) := (2, -2(1 + \log \ell))$ and target $L = \log T > 0$: the phase is $X(2 \log X - 2 - 2 \log \ell) = 2X \log(X/(e\ell))$, the proposition’s solution is $v_{\text{vol}} = W_0(\frac{L}{2}e^{-1-\log \ell}) = W_0(L/(2e\ell))$ and $X = \exp(v_{\text{vol}} + 1 + \log \ell) = e\ell e^{v_{\text{vol}}} = L/(2v_{\text{vol}}) = x$ (using $v_{\text{vol}}e^{v_{\text{vol}}} = L/(2e\ell)$), and its side condition $2 \log X + d_{\text{vol}} > 0$, that is $X > e\ell$, holds because $v_{\text{vol}} > 0$. Its slope identity reads $\Phi'_0(x) = 2(v_{\text{vol}} + 1) = 2h$: the report’s $h = \log(x/\ell)$ is $v_{\text{vol}} + 1$, and the derivative $2h$ used in Section 5 is the volume’s core slope.

- (b) X_1 and X_2 are an instance, after a change of variable, of `p0:thm:core-reversion`. Write $f(y)$ for the right side of (12) without its $O(n^{-1})$ term, with n replaced by a real y , and put $y = x(1 + E)$, $t_{\text{vol}} = x^{-1/2}$. Since $2x \log(x/(e\ell)) = L$ and $\log(x/(e\ell)) = h - 1$, the equation $f(y) = L$, divided by $2x$, is exactly the volume's master equation

$$hE + h_{\text{vol}}(E) + f_{\text{vol}}(t_{\text{vol}}, E) = 0, \quad h_{\text{vol}}(u) = (1 + u) \log(1 + u) - u,$$

$$f_{\text{vol}}(t, u) = \frac{\beta}{2}t(1 + u)^{1/2} - \frac{t^2}{8}(\log x + \log(1 + u)) + \frac{k}{2}t^2 + \frac{c_1}{2}t^3(1 + u)^{-1/2},$$

with $\Lambda_{\text{vol}} = h$, $h_{\text{vol}} \in u^2R[[u]]$ and $f_{\text{vol}} \in tR[[t, u]]$ over the ring $R = \mathbb{Q}(\beta, k, c_1, \log \ell)[\log x, h^{-1}]$, in which $\log x$ and $h = \log x - \log \ell$ are treated as constants, as `p0:rem:core-instances` does with the core slope. Its unique solution $E = \sum_j e_j t_{\text{vol}}^j$ has $xe_1 t_{\text{vol}} = -\beta\sqrt{x}/(2h)$, $xe_2 t_{\text{vol}}^2 = D_1$ and $xe_3 t_{\text{vol}}^3 = D_2$, so $X_1 = x(1 + e_1 t_{\text{vol}} + e_2 t_{\text{vol}}^2)$ and $X_2 = X_1 + xe_3 t_{\text{vol}}^3$ (the write checked the three coefficients symbolically; they are the source's three cancellation identities). The theorem is formal: the location errors $O((\sqrt{x}h)^{-1})$ and $O((xh)^{-1})$ are the source's own estimates.

- (c) $N(T)$ is the staircase N_* of `p0:def:three-inverses` for $A_n = a_n$ with $n_0 = n_1 = 0$, because the doubling injection makes a_n strictly increasing from $n = 0$ on. *The brackets (13) and the all-orders bracket are not instances of `p0:thm:staircase`, only analogues.* No admissible interpolation is constructed, and no bound $|x_* - \nu(T)| \leq \eta$ for one is proved: the brackets come from two envelopes, every integer left of the bracket having $a_n < T$ and its right end having $a_n \geq T$. For any admissible interpolation, `p0:thm:staircase(1)` gives $N(T) = \lceil \nu(T) \rceil$, which lies in the bracket, while $\nu(T)$ itself can be anywhere in $(N(T) - 1, N(T)]$. The source's warning that $\lceil X_1 \rceil$ alone is not justified is the failure mode named in `p0:thm:staircase(2)` and (5).
- (d) *The root y_M is not a formal object.* It is a real root of the smooth function Φ_M , found by Newton iteration from x ; neither `p0:thm:lambert-centered` nor `plt:thm:lw-template` produces it. `plt:thm:lw-template` does not apply to Φ_M : in the logarithmic chart $Z = \log y$ the term $\beta\sqrt{y}$ contributes terms in $e^{-Z/2}$ to its B , and the term $-\frac{5}{4}\log y$, together with Stirling's corrections, terms in Ze^{-Z} and e^{-Z} , all outside $tK[L_{\text{vol}}][[t]]$, so hypothesis (H3) fails. (Completed after the independent check of 6 October 2026; the first wording named only the term $\beta\sqrt{y}$, which already suffices.) It does apply to the core alone: $2y \log(y/(e\ell)) = L$ reads $Z + \log Z + \log 2 + \log(1 - (1 + \log \ell)/Z) = \log L$, a datum $(\rho, \mu, \beta_{\text{vol}}, B) = (1, 1, \log 2, \log(1 - (1 + \log \ell)t))$ with $B \in t\mathbb{R}[[t]]$, which gives the asymptotic series of x ; the source uses W_0 directly instead.

No novelty is claimed for these inversions; the Newton step is the generic one.

6 Exact enumeration and finite checks

Put $B_0 = B_1 = 1$ and

$$B_{k+1} = (2k + 1)B_k - k(k - 1)B_{k-1} \quad (k \geq 1). \quad (14)$$

Indeed, if $b(z) = \exp(z/(1 - z))$, then $(1 - z)^2 b'(z) = b(z)$, which gives (14) for $B_k = k![z^k]b(z)$. These are the numbers of sets of nonempty lists, OEIS A000262 [4]. Expanding (1) in powers of $(e^x - 1)(e^y - 1)$ and using

$$n![x^n]e^x(e^x - 1)^k = k!S(n + 1, k + 1)$$

gives the finite exact sum

$$a_n = \sum_{k=0}^n k! B_k S(n+1, k+1)^2. \quad (15)$$

Here $S(n, k)$ denotes a Stirling number of the second kind, defined by $S(0, 0) = 1$ and $S(n+1, k) = kS(n, k) + S(n, k-1)$, with zero values outside $0 \leq k \leq n$.

For R_n equal to the exact value divided by the leading term of (2), representative diagnostics are

n	R_n	$\sqrt{n}(R_n - 1)$	$n(R_n - 1 - c_1/\sqrt{n})$
100	0.9450185161	-0.5498148394	-0.1810850157
400	0.9729784069	-0.5404318613	-0.1745104690
800	0.9809866090	-0.5377799072	-0.1717864824
3200	0.9905481783	-0.5346757802	-0.1679770262

The limits of the last two columns are c_1 and c_2 . These finite computations are consistency checks, not error certificates for the asymptotic expansion or its onset.

Remark 4 ([write] The three OEIS entries, 6 October 2026). The live entries were fetched on 6 October 2026 (OEIS internal format); OEIS text is quoted verbatim and is CC BY-SA 4.0.

(a) *Names and data.* A299907 (revision #15, 25 October 2019) is named

Number of decomposable lonesum $n \times n$ $(0,1)$ matrices.

and its 16 terms ($0 \leq n \leq 15$) are those of `data/fixtures.json` and of (15). A299906 (revision #24, 26 October 2019) is named

Array read by antidiagonals: $T(n, k)$ = number of $n \times k$ lonesum decomposable $(0,1)$ matrices.

and its 66 terms are the antidiagonals of $D(m, n) = \sum_k k! B_k S(m+1, k+1) S(n+1, k+1)$. A000262 (revision #515, 16 August 2026) is named

Number of "sets of lists": number of partitions of $\{1, \dots, n\}$ into any number of lists, where a list means an ordered subset.

its first two formula lines read

D-finite with recurrence: $a(n) = (2n-1)a(n-1) - (n-1)(n-2)a(n-2)$.
E.g.f.: $\exp(x/(1-x))$.

which are (14) with $k = n - 1$ and the function b of Section 6, and its 22 data terms satisfy (14). The word order of A299907's name invites a false reading: "lonesum matrices that are decomposable" would be a subset of the lonesum matrices, of which there are 14 of size 2×2 (all but the two permutation matrices, by Kamano's Proposition 1.1), whereas $a_2 = 16$ counts every 2×2 matrix. The data are those of Kamano's lonesum decomposable matrices, the term A299906 uses (its name was corrected in 2019, by its own record).

(b) *The Mathematica line.* A299907 carries

```
a[n_] := Sum[Binomial[j - 1, k - 1]*j!^2*StirlingS2[n + 1, j + 1]^2/k!, {k, 0, n}, {j, k, n}]; Table[a[n], {n, 0, 15}] (* _Jean-François Alcover_, Feb 24 2018 *)
```

This is (15). Exchange the sums. Mathematica gives $\binom{-1}{-1} = 1$ and $\binom{j-1}{-1} = 0$ for $j \geq 1$ (checked by the write), so the terms with $k = 0$ contribute $S(n+1, 1)^2 = 1$, the term $j = 0$ of (15). For $j \geq 1$ the inner sum is $j! \sum_{k=1}^j \binom{j-1}{k-1} \frac{j!}{k!} = j! \sum_k L(j, k)$, with the Lah numbers $L(j, k) = \binom{j-1}{k-1} j! / k!$, and

$\sum_k L(j, k) = B_j$ because $\sum_k \frac{1}{k!} \left(\frac{z}{1-z}\right)^k = \exp \frac{z}{1-z} = b(z)$ and $\frac{1}{k!} \left(\frac{z}{1-z}\right)^k = \sum_j L(j, k) \frac{z^j}{j!}$ (the column generating function formalized as `Fabius.X_mul_mkOne_pow`). Before summing over k the line is Kamano’s Proposition 4.1, the count $D_k(n, n)$ of order k (with $D_0(n, n) = 1$, the zero matrix); so (15) is his formula summed over the order, the second route the source gives. The write evaluated the line in Wolfram 15: it returns the 16 data terms.

(c) *A misprint in A299906’s example.* The example table of A299906 prints its row $n = 5$ as

$$1, 32, 634, 8528, 90446, 833432, \dots,$$

so $T(5, 4) = 90446$. The correct value is 90946. Transposition is a bijection between 5×4 and 4×5 lonesum decomposable matrices (the definition is symmetric in rows and columns), so $T(5, 4) = T(4, 5)$, which the same table prints as 90946 in its row $n = 4$; the entry’s data list 90946 in both places; and $D(5, 4) = 1 + 1 \cdot 1 \cdot 31 \cdot 15 + 2 \cdot 3 \cdot 90 \cdot 25 + 6 \cdot 13 \cdot 65 \cdot 10 + 24 \cdot 73 \cdot 15 \cdot 1 = 1 + 465 + 13500 + 50700 + 26280 = 90946$ from the rectangular form of (15) (the factors are $k!$, B_k , $S(6, k + 1)$ and $S(5, k + 1)$ for $k = 0, \dots, 4$). Recorded here only; nothing was submitted to the OEIS.

Reproducible calculations. The accompanying bundle includes the article source, a deterministic PDF build, integer enumeration, and symbolic coefficient extraction. The script `derive_general.py` computes the recurrence (10) and Gaussian moments using $\ell = 8z^2$ to avoid nested radicals. It accepts every nonnegative coefficient order, including zero. The separate script `derive.py` obtains P_1, P_2, c_1 by direct expansion with ℓ . The default finite checker verifies exact small fixtures, exhaustive forbidden-submatrix counts through $n = 3$, the symbolic formulas for c_1, c_2 , parity, and the two displayed inverse cancellations. All checks use explicit guards and also run under Python’s optimization flag. Optional high- n diagnostics are separate so routine checking does not require the $n = 3200$ computation.

The bundle’s integrity manifest covers a closed file inventory. Its quality-assurance procedure runs checks on immutable copies, rejects selected modified fixtures and formulas after rebuilding their integrity manifest, compares two clean PDF builds byte for byte, and replays a freshly extracted archive. These tests check reproducibility and the finite identities they name. They do not mechanize or replace the analytic proof in Sections 2–4.

[write] **The package here, and the intake reruns, 6 October 2026.** “The accompanying bundle” above is the delivered one. In this report the programs `derive_general.py`, `derive.py`, `exact.py`, `formulas.py`, `validate.py`, `check.py`, `build.py`, `qa.py`, `package.py` and `safe_output.py` are in `code/`; `fixtures.json`, `diagnostics_3200.json` and `requirements.txt` are in `data/`; `sources.md` is at the report root; the article source is `article.tex`. The delivered PDF and the integrity manifest `MANIFEST.json` are not shipped. The programs import one another and read `fixtures.json` and the article from their own directory, and `check.py` first verifies a closed inventory of the 18 delivered names against the manifest, so its default mode runs only in a fresh extraction of the arrival archive; on a flat copy of the shipped programs, the fixtures and the article (renamed `Report132.tex`) its `-finite-only` mode runs every finite check (the report README gives both routes). On copies at the write (Windows, Python 3.14.4, SymPy 1.14.0, mpmath 1.3.0): in a fresh extraction `check.py` passed in normal and `-O` mode (about five seconds), `validate.py--max-n100` reproduced the row $n = 100$ of the table, and `derive_general.py3` gave c_3 (Remark 2); on the flat copy, `-finite-only` passed and the default mode stopped at its inventory check, as designed. The PDF build, `qa.py` and `package.py` need the whole delivered bundle and a POSIX TeX installation and were not run.

[write] **Independent check of the write, 6 October 2026.** An adversarial check made by the intake after the write (commit 6ef3f0983) re-derived the statements marked [write] with its own code. *Remark 1.* Every $m \times n$ binary matrix with $m, n \leq 4$ was classified by four independent deciders: the literal definition (a search for a block decomposition, with lonesum tested as uniqueness of the pair of line-sum vectors, without Ryser’s criterion), criterion (b), Kamano’s twelve forbidden matrices generated from U , and a search for induced paths

on five vertices; every $m \times n$ matrix with $m, n \leq 5$ (all 2^{25} at 5×5) was classified by the last two. There was no pointwise disagreement; every count equals the rectangular form of (15), e.g. $D(5, 4) = D(4, 5) = 90946$ and $D(5, 5) = 833432$; and through 5×5 the lonesum matrices without zero lines are exactly the connected chain graphs. Kamano's set and his Theorem 2.1 were read again in the arXiv text. *Remark 2.* An independent truncated trivariate expansion of (9) from its definition, through t^8 at 50 and 90 digits, reproduced β , γ , A , B , C , the vanishing odd orders, and c_1 , c_2 (differences $2 \cdot 10^{-91}$ and $9 \cdot 10^{-90}$ from (4) and (5)); its c_3 agrees with the closed form of the remark to 88 digits, and it gives $c_4 = 1.003420176595018111637014920569 \dots$; rerun when the check was applied, the delivered `derive_general.py` at order 4 (about fifteen seconds) gives the same c_4 to 69 digits, as a rational function of ℓ with denominator $30 \cdot 192^4 \ell^6 (\ell - 1)^4$ and a numerator of degree 12 (the pattern of Question 5; not claimed). Exact a_n for $n = 100, 200, 400, 800, 1600, 3200$ reproduce every digit of the table above and every number of the remark; $n^2(R_n - \sum_{j \leq 3} c_j n^{-j/2})$ is 0.916, 0.959, 0.972, 0.981, 0.987 at $n = 100, 400, 800, 1600, 3200$, approaching c_4 , and with the c_4 term also subtracted the remainder times $n^{5/2}$ is $-0.874, \dots, -0.907$, stable. *Remark 3.* The seven statements of the volume that it uses were read again; SymPy confirms that $f(y) = L$ divided by $2x$ is identically the master equation of (b) and that its recurrence gives $-\beta/(2h)$, D_1 and D_2 ; (a), (c) and (d) were re-derived. *Remark 4.* In the live A299906 (still revision #24) the example row $n = 5$ prints 90446, while row $n = 4$ and the data print 90946; $T(5, 4) = 90946$ by the formula, by enumeration of all 2^{20} matrices of size 5×4 and by transposition, and all 66 data terms are antidiagonals of $D(m, n)$. The 16 terms of A299907 equal (15), and a_n for $n \leq 5$ also follows from the definition by enumeration; the names, revisions and the quoted formula lines of A000262 match. No mathematical error was found. Two statements are made more precise, each with a dated note keeping the first wording: Remark 2 now states c_4 , and Remark 3(d) names all the terms that violate (H3).

7 Further questions

1. **Effective remainders.** The proof gives constants and onset thresholds that exist for every fixed order. Explicit numerical values would turn the inverse brackets into certified finite- T tools; this requires quantitative bounds throughout the contour argument.
2. **Rectangular asymptotics.** For m/n tending to a fixed positive ratio, the saddle approaches a generally nonsymmetric point on $(e^x - 1)(e^y - 1) = 1$. Uniform expansions as the aspect ratio varies, especially toward an unbalanced regime, remain a separate problem.
3. **The number of blocks.** Kamano's marked generating function is

$$F(x, y; w) = \exp \left(x + y + w \left(\frac{1}{1 - (e^x - 1)(e^y - 1)} - 1 \right) \right).$$

A uniform analysis in w near 1 should be a starting point for the distribution of the decomposition order. No central or local limit theorem is asserted here.

4. **Beyond the power series.** The expansion concerns every fixed algebraic order relative to the leading term. Optimal truncation and exponentially small terms require additional analysis; the present proof does not identify them.

[write] **Added 6 October 2026, batch 108.** Following Vladimir Reshetnikov's standing rule of 4 October 2026, every claim of the source that is not proved in full is stated as an open question with its source, the argument offered and what is missing. The four questions above are the source's own, kept as delivered; its non-claims are collected in Section 1.1. Two statements of the source made without proof are now proved (Remark 1: each nontrivial component is a chain graph, and the forbidden configurations are *precisely* the induced P_5), and the stored but unclaimed c_3 is now an output of the delivered recipe (Remark 2). The intake found no false claim in the source and the write found none either; the one misprint corrected (Remark 4(c)) is in OEIS text. What each source question needs, as the write reads it: item 1 needs explicit bounds in

Lemma 1 (the constants c , η and the gap c_0 come from continuity and compactness), in the Taylor remainders of Section 4 and in the comparison of the explicit radius with the box, after which the brackets of Section 5 become checkable for given T ; item 2 needs the saddle on the curve $(e^x - 1)(e^y - 1) = 1$ uniformly in the aspect ratio, which A299906 would test numerically; item 3 starts from Kamano’s per-order generating function [1, Theorem 3.1, equation (8)], which is the coefficient of w^k in the marked function above; item 4 is untouched.

5. **The shape of c_j** (added by the write, not a claim of the source). For $j = 1, 2, 3$ the coefficient c_j is $\sqrt{2}$ (for odd j) or 1 (for even j) times a polynomial in ℓ of degree $3j$ with integer coefficients, divided by $\delta_j \ell^{3j/2} (\ell - 1)^j$ with $\delta_j = 192, 192^2, 15 \cdot 192^3$ (equations (4), (5) and Remark 2). Does this persist, and is there a closed form for the denominators, or a generating function for the c_j ? The recipe gives each c_j in finitely many operations, so the question is structural.
6. **The literature boundary** (from the source’s prior-art paragraph). Khera’s dissertation [7] treats lonesum matrices and their asymptotics; neither the source nor the write read it, and it is not known here whether it treats lonesum decomposable matrices. Reading it would settle whether the leading equivalent of Theorem 1 is already in print.

[write] **Reciprocal pointers, 6 October 2026.** The write proposes, without applying them, a pointer to this report in `a122399-surjection-diagonal` (the nearest neighbour; see Section 1.1) and a correction of the sentence “A000262 occurs nowhere else in the repository” in the README of `a321941-asymptotic-coefficient-integrality`, which this report makes stale.

References

- [1] K. Kamano, *Lonesum decomposable matrices*, Discrete Mathematics 341 (2018), 341–349. DOI: 10.1016/j.disc.2017.09.002. Preprint arXiv:1701.07157, Theorem 3.1, equation (9), PDF page 7.
- [2] OEIS Foundation Inc., A299907.
- [3] OEIS Foundation Inc., A299906, rectangular lonesum decomposable matrices.
- [4] OEIS Foundation Inc., A000262, sets of lists.
- [5] J. Khera, E. Lundberg and S. Melczer, *Asymptotic enumeration of lonesum matrices*, Advances in Applied Mathematics 123 (2021), 102118. doi:10.1016/j.aam.2020.102118; arXiv:1912.08850. See also Remark 2 on higher orders.
- [6] B. Bényi and J. L. Ramírez, *Some applications of S -restricted set partitions*, arXiv:1804.03949, Section 4.3 and Theorem 11.
- [7] J. Khera, *Lonesum Matrices and Acyclic Orientations: Enumeration and Asymptotics*, Ph.D. dissertation, Florida Atlantic University, 2021. University dissertation listing; defense abstract. Full text not inspected.